

Key:

- **Questions marked with a 🍷** are for discussion in your tutorials. Please complete these questions before your tutorial.
- **Questions marked with a 🦋** are difficult questions.
- **Questions marked with a 🎵** have been marked so that they don't feel left out. Please don't leave them out during your independent learning and revision time.

🍷 **Exercise 1.** Compute the limit $\lim_{x \rightarrow 0} \frac{1 - \sqrt{1 - x^2}}{x^2}$

🍷 **Exercise 2.** Given that $\lim_{x \rightarrow 0} \frac{\sin(x)}{x} = 1$, explain why $\lim_{x \rightarrow 0} \frac{\sin|x|}{x}$ does not exist.

🍷 **Exercise 3.** Determine the following limits.

(i) $\lim_{x \rightarrow 2} \frac{x^2 + 2x - 8}{x(x - 2)}$

(ii) $\lim_{x \rightarrow 1} \frac{x^3 - 1}{x - 1}$

(iii) $\lim_{x \rightarrow 0} \frac{x - \frac{1}{x}}{x + \frac{1}{x}}$

(iv) $\lim_{x \rightarrow 1} \frac{x^{27} - 1}{x - 1}$

🎵 **Exercise 4.** Compute the limit $\lim_{x \rightarrow \infty} x \left(\sqrt{x^2 + 3} - x \right)$.

🎵 **Exercise 5.** Sketch a possible graph of $f(x)$ in the following cases. Assume $f(x)$ is defined and continuous for all real x .

(i) $\lim_{x \rightarrow \infty} f(x) = -\infty$ and $\lim_{x \rightarrow -\infty} f(x) = \infty$.

(ii) $\lim_{x \rightarrow \infty} f(x) = 1$ and $\lim_{x \rightarrow -\infty} f(x) = \infty$.

(iii) $\lim_{x \rightarrow 3} f(x) = 5$ and $\lim_{x \rightarrow \pm\infty} f(x) = \infty$.

🦋 **Exercise 6.** Compute the limit

$$\lim_{x \rightarrow \infty} x^{2/3} \left((x+1)^{1/3} - (x-1)^{1/3} \right).$$

✦ **Exercise 7.** Using the formal definition of a limit, prove that:

(i) $\lim_{x \rightarrow 0} x^2 = 0$;

(ii) $\lim_{x \rightarrow 0} (x^5 + x) = 0$;

(iii) The function $f(x) = \cos\left(\frac{1}{x^2}\right)$ has no limit as $x \rightarrow 0$. Here, the domain of f is $\mathbb{R} \setminus \{0\}$.